

Program B: Minimum Distance Between Two Numbers

Question

Given an unsorted array of integers and two numbers **a** and **b**, design an algorithm and write a C program to find the minimum distance between **a** and **b**. The minimum distance is the minimum difference between their indices.

Algorithm

1. Read the size **n**, the array elements, and the values **a** and **b**.
2. Set **previousIndex = -1** and **minimumDistance = n + 1**.
3. If **a = b**, calculate the distance between consecutive occurrences of that value.
4. Otherwise, traverse the array and process every occurrence of **a** or **b**.
5. When the current target differs from the previously found target, calculate their index difference.
6. Update **minimumDistance** whenever a smaller distance is obtained.
7. Display the minimum distance; if no valid pair exists, display an appropriate message.

C Code

```
#include <stdio.h>

#define MAX_SIZE 100

int main(void)
{
    int array[MAX_SIZE];
    int n, a, b, i, previousIndex = -1;
    int distance, minimumDistance;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    if (n <= 0 || n > MAX_SIZE) {
        printf("Invalid array size.\n");
        return 1;
    }

    printf("Enter %d array elements:\n", n);
    for (i = 0; i < n; i++)
        scanf("%d", &array[i]);

    printf("Enter a and b: ");
    scanf("%d %d", &a, &b);

    minimumDistance = n + 1;

    if (a == b) {
        for (i = 0; i < n; i++) {
            if (array[i] == a) {
                if (previousIndex != -1) {
                    distance = i - previousIndex;
                    if (distance < minimumDistance)
                        minimumDistance = distance;
                }
                previousIndex = i;
            }
        }
    } else {
        for (i = 0; i < n; i++) {
            if (array[i] == a || array[i] == b) {
                if (previousIndex != -1 &&
                    array[i] != array[previousIndex]) {
                    distance = i - previousIndex;
                    if (distance < minimumDistance)
                        minimumDistance = distance;
                }
                previousIndex = i;
            }
        }
    }

    if (minimumDistance == n + 1)
        printf("A valid pair was not found.\n");
    else
        printf("Minimum distance between %d and %d = %d\n",
            a, b, minimumDistance);

    return 0;
}
```